

PROJECT PLANNING PHASE

ComicCraft – AI Comic Story Creator using Gemini Models

1. Introduction to Project Planning

The Project Planning Phase defines how the ComicCraft application will be developed, organized, tested, and prepared for local deployment. ComicCraft is a web-based application that uses AI to generate personalized comic stories and illustrations from user-provided prompts. The project document divides the overall work into milestones covering model selection and architecture, core functionality development, routes.py development, frontend development, and deployment.

2. Main Objectives

- Select suitable generative AI models for story and image generation.
- Define the application architecture and responsibilities of each component.
- Prepare the Python development environment and required libraries.
- Plan the core comic-generation workflow from user input to PDF export.
- Plan backend routes and frontend templates for smooth user interaction.
- Prepare and verify the application for local deployment.

3. Planning Approach

The project is planned as a sequence of connected activities. User input is collected through the frontend, processed by the FastAPI backend, passed to AI models, organized into comic panels, and finally exported as a PDF. This planned workflow provides a clear path from the initial idea to a usable application.

4. Project Milestones and Activities

Milestone 1: Model Selection and Architecture

The first milestone focuses on researching and selecting appropriate generative AI models and defining the architecture of ComicCraft. The project document identifies Gemini Flash for structured comic outlines, Gemini Pro for detailed narration and dialogues, and Stable Diffusion for comic-style illustrations.

Planned Activities

- Activity 1.1 – Research and select the appropriate generative AI model.
- Activity 1.2 – Define the architecture of the application.
- Activity 1.3 – Set up the development environment.

Milestone 2: Core Functionalities Development

The second milestone plans the main AI-powered functions required to create a complete comic. The workflow includes generating a structured five-panel outline, expanding it into narration and dialogue, creating illustrations, organizing panels, and exporting the final comic as a PDF.

Planned Activities

- Activity 2.1 – Develop the core functionalities.
- Activity 2.2 – Implement the FastAPI backend to manage routing and user input processing.

Key Planned Functions

`generate_outline()` – Creates the structured comic panel outline.

`generate_story()` – Creates narration and character dialogues.

`generate_image()` – Generates comic-style illustrations.

`build_comic_layout()` – Organizes images and story content into a panel layout.

`save_pdf()` – Compiles the comic into a multi-page PDF.

5. Backend and Route Planning

Milestone 3: routes.py Development

The third milestone plans the main application logic in routes.py. Routes act as entry points for different user interactions and connect the frontend with the comic-generation workflow.

Core Routes Planned

Route	Purpose
/	Loads the homepage and user input form.
/generate	Processes form input and generates the comic preview.
/generate-comic/json	Accepts JSON input and returns comic layout data and PDF path.
/export-success	Displays confirmation after comic export.
/test-image	Tests image generation from a direct prompt.

6. User Input Planning

The planning includes five main user inputs: story prompt, main character name, setting, story tone, and art style. These values are captured through the HTML form and passed to the FastAPI backend. For API-based generation, a Pydantic PromptRequest schema is planned for JSON validation.

7. Error Handling Plan

The route handlers are planned to use FastAPI's HTTPException system and try-except blocks so that unexpected API errors or JSON decoding problems can be handled gracefully.

8. Frontend, Environment and Deployment Planning

Frontend Development Planning

The frontend plan uses HTML, CSS, and Jinja2 templates. The application is divided into separate templates for collecting input, displaying the generated comic, and confirming successful export.

Planned Templates

- `index.html` – Collects story prompt, character name, setting, tone, and art style.
- `comic_preview.html` – Displays generated comic panels with images and narration.
- `export_success.html` – Displays confirmation after the comic is downloaded.

Development Environment Plan

The project document plans a Python virtual environment to isolate project dependencies. Required packages include FastAPI, Uvicorn, Jinja2, python-multipart, Google Generative AI, Diffusers, Transformers, FPDF, Pillow, and Accelerate.

Environment and Security Planning

Sensitive credentials such as the Gemini API key and Hugging Face API token are planned to be stored using environment variables or a `.env` file loaded through `python-dotenv`.

Local Deployment Plan

The application is planned for local deployment using Uvicorn. The project document specifies launching the FastAPI application with the command **`uvicorn app.main:app --reload`**. The application can then be accessed locally through the browser, while FastAPI's `/docs` page can be used for API testing.

9. Project Planning Summary

Planned End-to-End Workflow

The planned ComicCraft workflow connects all project components in the following order:

1. User enters the story prompt and comic preferences.
2. FastAPI receives and processes the user input.
3. Gemini Flash creates a structured five-panel outline.
4. Gemini Pro generates narration and character dialogues.
5. Stable Diffusion generates an illustration for each panel.
6. The generated images and story content are organized into a comic layout.
7. FPDF is used to create the final multi-page PDF.
8. The user receives the comic preview and can download the completed PDF.

10. Expected Project Outcome

The planning phase is intended to ensure that ComicCraft has a clear development structure and that each component has a defined role. The project document describes an end-to-end system capable of turning a simple creative prompt into a personalized comic with story, dialogue, illustrations, and a downloadable PDF.

11. Conclusion

The Project Planning Phase provides the roadmap for implementing ComicCraft. By dividing the work into milestones and activities, the project establishes a structured path for AI model integration, backend development, frontend development, testing, and local deployment. This planning supports the overall goal of creating an intuitive AI-powered comic-generation application.

Source: ComicCraft – AI Comic Story Creator using Gemini Models project document.